

DOES GOOD INFRASTRUCTURE HAVE AN EVEN BIGGER EFFECT ON GDP IN DEMOCRACIES THAN IN AUTOCRACIES

DATA SCIENCE MA705-01: FINAL PROJECT

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EXECUTIVE SUMMARY

This study examined whether the effect of infrastructure quality on GDP differs between democracies and autocracies. Using merged data from the 2023 Legatum Prosperity Index and World Bank GDP figures, we applied multiple linear regression models with an interaction term between infrastructure and regime type. The initial model explained only 10.4% of GDP variation, and the interaction effect was not statistically significant. However, after removing two influential outlier countries with extreme GDP values, the adjusted model improved significantly, explaining 23.1% of GDP variation. Most importantly, the interaction term became statistically significant ($p = 0.009$), indicating that the relationship between infrastructure and GDP varies meaningfully across regime types.

The key finding was that infrastructure improvements were linked to larger GDP decreases in democracies than in autocracies. While autocratic countries showed modest declines in GDP as infrastructure scores increased, the effect in democracies was more negative. A supporting visualization confirmed this result, with a steeper downward slope for democracies. This suggests that infrastructure investments may be less economically effective in democratic regimes, potentially due to challenges like slower decision-making or implementation inefficiencies. These findings highlight that infrastructure alone does not guarantee economic growth. The political environment plays a critical role in shaping outcomes, and in democracies especially, complementary reforms may be necessary to ensure infrastructure spending leads to meaningful economic gains.

INTRODUCTION

Problem Statement

Infrastructure is widely recognized as a driver of economic growth, but its impact may not be consistent across political systems. This project investigates whether the effect of infrastructure

quality on GDP is greater in democracies than in autocracies. Understanding this interaction can help guide infrastructure investment decisions based on political context.

Relevance of the Study

While infrastructure is essential for productivity and development, its success depends on how it is implemented. Democracies may benefit from stronger institutions and accountability, while autocracies may face challenges like inefficiencies or misallocation. This study explores how political systems shape the effectiveness of infrastructure investments, offering insights for policymakers, investors, and development agencies.

Objectives

The goal of this project is to analyze the relationship between infrastructure quality and GDP across countries, and to test whether this relationship is influenced by regime type. Using statistical models with interaction terms, we assess whether democracies derive greater economic benefit from infrastructure than autocracies. The findings aim to support more targeted and effective infrastructure planning in different governance contexts.

THE DATASETS

Sources of Data

This project combines data from two primary sources. The first dataset is the *2023 Global Prosperity Index* compiled by the Legatum Institute and accessed via Kaggle. It includes country-level scores for various prosperity sub-factors such as infrastructure, governance, health, and education, along with regional and political regime classifications.

The second dataset comes from the World Bank and provides 2023 GDP figures for each country, measured in current US dollars. These two datasets were merged to facilitate a comparative analysis of how infrastructure quality and political regime types relate to economic performance.

Structure of the Datasets

The Prosperity Index dataset contains 159 countries, with variables such as country name, region, political regime, overall prosperity score, and sub-factors like governance, infrastructure, health, and education. Most variables are numeric, except for region and regime type.

The World Bank GDP dataset includes GDP values for 2023 in current US dollars, along with country names, ISO codes, and indicator metadata. After filtering, both datasets were merged on country name to allow for integrated analysis of GDP, infrastructure, and political systems.

DATA WRANGLING

To prepare the data for analysis, we began by importing two separate datasets. One provided global prosperity indicators, while the other contained GDP values for countries in the year 2023. From the GDP dataset, we selected only the columns relevant to our analysis, allowing us to maintain a focused and manageable structure. During this stage, we also removed duplicate fields that resulted from merging, such as redundant entries for country names and codes. To ensure consistency throughout our analysis, we standardized all column names. Several prosperity indicators, including variables like education, health, and infrastructure, were originally recorded as decimal values. Because these scores reflect ranked indices, we converted them into whole numbers to improve clarity and interpretability. Additionally, we removed countries with missing GDP values to maintain the completeness and reliability of the dataset used for modelling.

One of the most important parts of the processing phase was aligning country names across both datasets. We checked for mismatches using a comparison function and created a correction map to manually fix naming inconsistencies. This ensured that the merge operation only included valid and matching records. Once corrections were applied, we successfully merged the datasets using an inner join, retaining only countries with complete data in both sources. We then renamed key columns, such as updating "2023" to "gdp_2023," to make their content more immediately understandable. Finally, we created a new column that grouped each country's political system as either a "Democracy" or "Autocracy," setting the foundation for the regression and comparative analysis that followed. These steps ensured the dataset was clean, consistent, and fully prepared for accurate statistical modelling.

An **inner join** was used to merge the two datasets based on the shared country code and name. This approach ensured that only countries present in both datasets were retained, allowing for complete and valid values across all columns. After cleaning for missing values in the 2023 GDP column, 152 out of 159 countries from the Prosperity Index remained in the final dataset. Since the Prosperity dataset had fewer entries initially, most of its rows were preserved. The merged dataset contains 152 rows and 23 columns, reflecting the combined fields from both sources, along with one additional column created during data wrangling. Merge issues were minimal, and no missing values remained post-merge.

METHODOLOGY

To examine the relationship between national characteristics and economic performance, multiple linear regression was used to model how various sub-factors such as governance, education, health, infrastructure, social capital, economic quality, and living conditions relate to a country's GDP in 2023. Those predictors were selected particularly to account for other important factors that could significantly influence the GDP based on the merged dataset. Multiple linear regression allowed us to control for multiple variables simultaneously, helping to

isolate the individual contribution of each factor while accounting for the broader policy and social context within each country.

An interaction term between infrastructure quality and regime type (classified as democracy or autocracy) was introduced to explore whether the influence of infrastructure on GDP differs by political regime. This method allows for the testing of moderating effects; whether the relationship between infrastructure and GDP changes depending on the governing system. By including this interaction, the model could capture not just whether infrastructure is important, but whether its impact is stronger or weaker in democracies compared to autocracies.

These methods were chosen for their ability to quantify the relative importance of various governance-related factors while also detecting subtle differences in how political structures might shape economic outcomes. In doing so, the analysis moves beyond surface-level associations to uncover deeper patterns of influence between infrastructure, political regime, and economic performance. Two models were run to understand the relationship.

RESULTS

Model 1: Initial Model (Unadjusted for Outliers)

In the first model, we tested whether the relationship between infrastructure and GDP differs across regime types using select predictor variables. This model included an interaction between *infrastructure quality* and *regime type* (Democracy vs. Autocracy), while also accounting for other factors that influence GDP: education, health, living conditions, social capital, and overall economic quality.

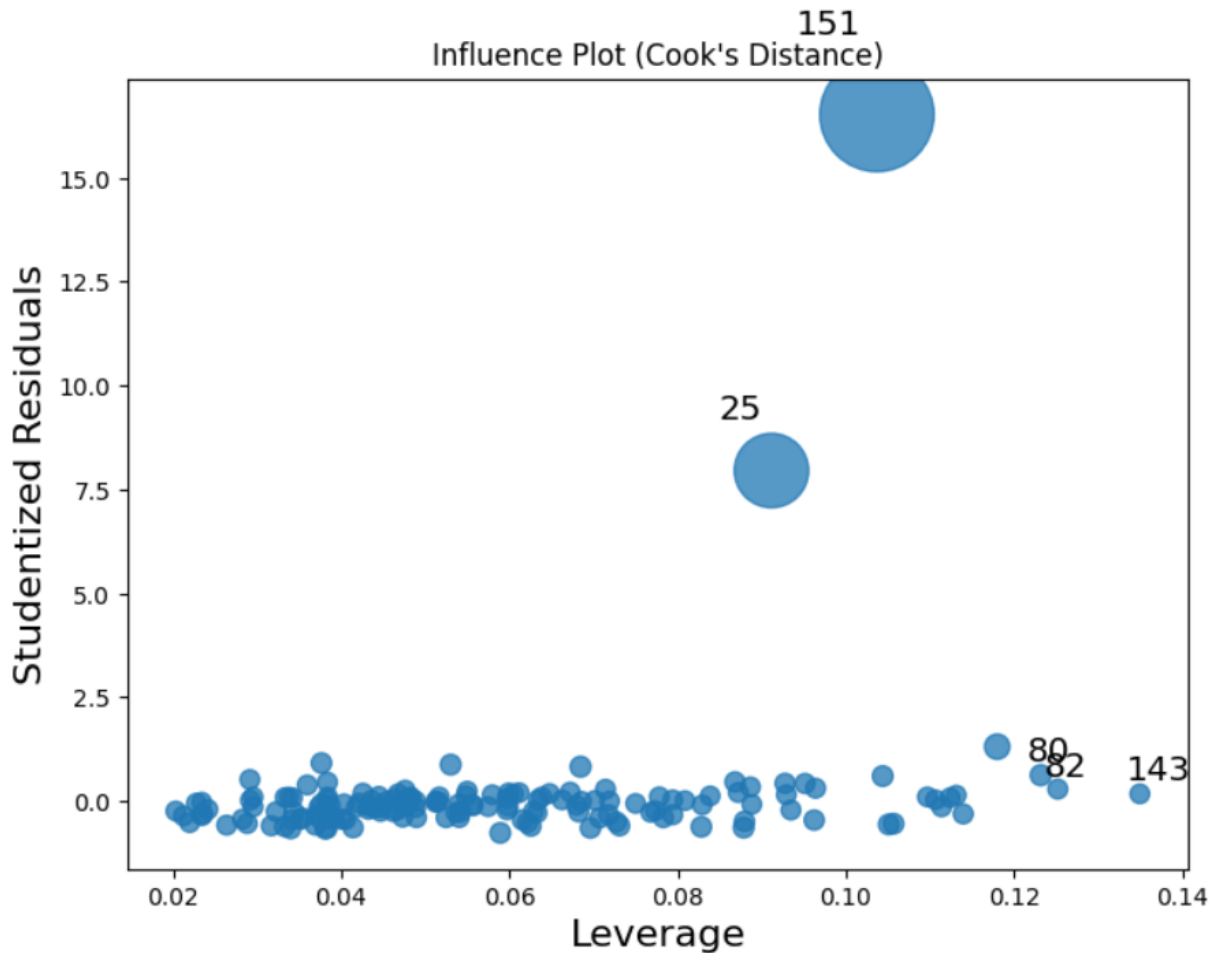
```
model = smf.ols('gdp_2023 ~ infrastructure_and_market_access * Regime_Group +  
education + health + economic_quality + living_conditions + social_capital',  
data=df_merged). fit()
```

OLS Regression Results						
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Dep. Variable:	gdp_2023	R-squared:	0.104			
Model:	OLS	Adj. R-squared:	0.054			
Method:	Least Squares	F-statistic:	2.083			
Date:	Sun, 27 Apr 2025	Prob (F-statistic):	0.0412			
Time:	21:46:52	Log-Likelihood:	-4559.6			
No. Observations:	152	AIC:	9137.			
Df Residuals:	143	BIC:	9164.			
Df Model:	8					
Covariance Type:	nonrobust					
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		coef	std err	t	P> t	[0.025 0.975]

Intercept		1.88e+12	9.04e+11	2.079	0.039	9.26e+10 3.67e+12
Regime_Group[T.Democracy]		4.025e+11	1.01e+12	0.398	0.691	-1.6e+12 2.4e+12
infrastructure_and_market_access		-2.766e+10	1.56e+10	-1.770	0.079	-5.86e+10 3.23e+09
infrastructure_and_market_access:Regime_Group[T.Democracy]		-1.173e+10	1.07e+10	-1.094	0.276	-3.29e+10 9.46e+09
education		-1.387e+09	1.49e+10	-0.093	0.926	-3.08e+10 2.8e+10
health		1.369e+10	1.22e+10	1.124	0.263	-1.04e+10 3.78e+10
economic_quality		-3.708e+09	1.15e+10	-0.322	0.748	-2.64e+10 1.9e+10
living_conditions		1.365e+10	1.62e+10	0.842	0.401	-1.84e+10 4.57e+10
social_capital		-8.193e+09	6.64e+09	-1.233	0.220	-2.13e+10 4.94e+09

Looking at the summary of the model, the model explained about 10.4% of the variation in GDP across countries which is small but understandable as it simply reflects the complexity of real-world data at the national level. The interaction term (infrastructure $\times$ regime type) was not statistically significant ($p = 0.276$), meaning the data did not provide strong evidence that the effect of infrastructure differs between democracies and autocracies. This implies that there is no strong statistical evidence that infrastructure affects GDP differently between democracies and autocracies after controlling for education, health, economic quality, living conditions, and social capital. Although the coefficient is negative (suggesting a slightly smaller impact in democracies), the difference is not statistically meaningful.

To understand how points could be distorting the regression coefficients, an influence plot was made. The influence plot helps identify observations that disproportionately affect the regression line or model fit. The influence plot shows three things: Studentized Residuals (Y-axis): How extreme the residual is (how far it is from the predicted value, standardized). The X-axis shows how extreme the predictor values are (how far an observation's X-values are from the mean of X). Lastly, the size of the points shows level of influence.



After running the diagnostic plot, it revealed that two countries were exerting an unusually large influence on the model due to their extreme GDP values and prediction errors. This prompted the creating of a second model that accounts for outliers. The two countries were removed to create the second model.

Model 2: Final Model (After Removing Outliers)

```
model_clean = smf.ols('gdp_2023 ~ infrastructure_and_market_access * Regime_Group +  
education + health + economic_quality + living_conditions + social_capital',  
data=df_clean). fit()
```

To improve the reliability of the model, the two outlier countries in the influence plot (151 and 25) were removed and the analysis was rerun. The adjusted model's performance improved significantly; it now explains 23.1% of GDP variation relative to the 10.4% R-square of the initial model. Most importantly, the interaction term became statistically significant ($p = 0.009$). The results showed that for autocracies, improvements in infrastructure were associated with modest decreases in GDP. However, in democracies, the same improvements were associated

with even larger decreases in GDP. This means that the effect of infrastructure on GDP is not the same across regime types. Somewhat surprisingly, infrastructure improvements had a stronger negative association with GDP in democracies than in autocracies in this dataset.

OLS Regression Results						
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Dep. Variable:	gdp_2023	R-squared:	0.231			
Model:	OLS	Adj. R-squared:	0.187			
Method:	Least Squares	F-statistic:	5.282			
Date:	Sun, 27 Apr 2025	Prob (F-statistic):	8.46e-06			
Time:	21:54:28	Log-Likelihood:	-4301.5			
No. Observations:	150	AIC:	8621.			
Df Residuals:	141	BIC:	8648.			
Df Model:	8					
Covariance Type:	nonrobust					
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	coef	std err	t	P> t	[0.025	0.975]

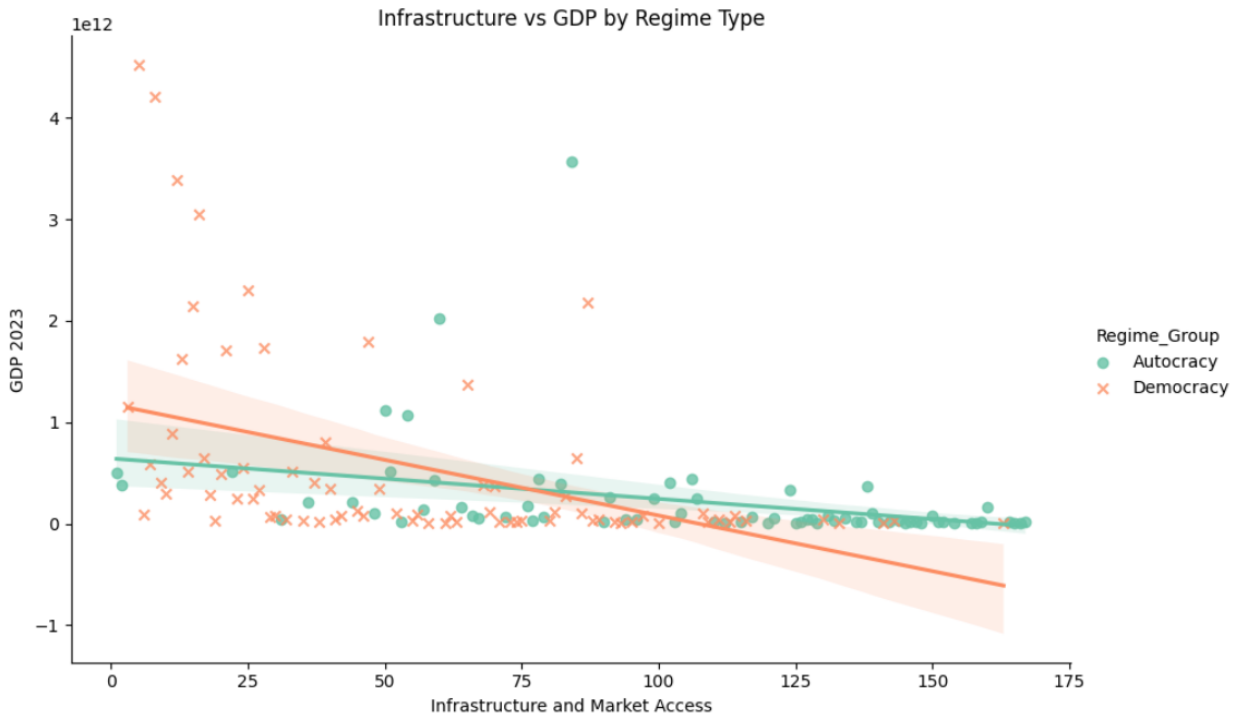
Intercept	5.397e+11	2.47e+11	2.183	0.031	5.1e+10	1.03e+12
Regime_Group[T.Democracy]	6.185e+11	2.75e+11	2.247	0.026	7.45e+10	1.16e+12
infrastructure_and_market_access	-7.584e+09	4.22e+09	-1.798	0.074	-1.59e+10	7.55e+08
infrastructure_and_market_access:Regime_Group[T.Democracy]	-7.707e+09	2.91e+09	-2.650	0.009	-1.35e+10	-1.96e+09
education	-1.973e+09	4e+09	-0.493	0.623	-9.88e+09	5.94e+09
health	-1.544e+08	3.4e+09	-0.045	0.964	-6.88e+09	6.57e+09
economic_quality	5.643e+08	3.08e+09	0.183	0.855	-5.52e+09	6.65e+09
living_conditions	5.339e+09	4.34e+09	1.231	0.220	-3.23e+09	1.39e+10
social_capital	7.991e+08	1.79e+09	0.447	0.655	-2.73e+09	4.33e+09
=====						

Visualization: Infrastructure vs GDP by Regime Type

The scatter plot below illustrates how infrastructure quality relates to GDP, with separate trend lines for democracies and autocracies. Countries governed by autocratic regimes are shown as green circles, while democratic countries are represented by orange crosses. The x-axis measures infrastructure and market access scores, and the y-axis shows GDP in 2023. The trend lines, along with their shaded confidence intervals, help visualize the overall direction and reliability of the relationships.

Interestingly, both regime types display a downward trend, indicating that higher infrastructure scores are generally associated with lower GDP in this dataset. This pattern is more pronounced in democracies, where the negative slope is steeper. While the result may seem counterintuitive, the plot provides a clear visual cue that the relationship between infrastructure and GDP is not consistent across political systems.

Rather than concluding that infrastructure is harmful, the chart reinforces the idea that its effectiveness may be shaped by political and institutional conditions. The visual separation between the two trend lines hints that the regime type may play a role in how infrastructure investments influence economic outcomes



REAL WORLD IMPLICATIONS

The results of the adjusted model suggest that infrastructure improvements do not boost GDP the same way in every country. The analysis shows that infrastructure is linked to larger gains in GDP for autocratic countries than for democratic ones. In democracies, the impact of infrastructure on the economy appears smaller and, in some cases, may even be negative.

This doesn't mean that infrastructure isn't valuable, but it does mean that politics and decision-making processes matter. In autocratic systems, governments can often move quickly and launch large infrastructure projects with little delay. In contrast, democracies may face more red tape, slower approval processes, and political disagreements that slow down or weaken the impact of these projects.

For leaders, investors, and international partners, this highlights an important takeaway: just spending money on infrastructure isn't enough. In democratic countries especially, it's important to also invest in strong systems, better coordination, and reducing delays; otherwise, the full benefits of infrastructure spending may not be realized.

Ultimately, this analysis shows that infrastructure can help grow economies, but how it's planned, delivered, and under what kind of government makes a real difference.

CONCLUSION

Our findings show that infrastructure has a stronger positive effect on GDP in autocratic countries than in democratic ones. After adjusting for outliers, the results revealed that improvements in infrastructure were linked to economic growth in autocracies, while the effect in democracies was weaker and even slightly negative. This goes against the expectation that democracies, with their stronger institutions and transparency, would see greater economic returns from infrastructure investments.

These results highlight the importance of political context in shaping economic outcomes. Future studies could explore the reasons behind this pattern by examining how project delivery, governance challenges, or long-term planning differ between regime types. It would also be valuable to study whether certain types of infrastructure, such as transport or digital systems, perform differently across political systems.

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